

2021 년 가을학기 (중간고사)

1. Find an extremal for the functional

$$J(x) = \int_1^{t_f} [2x(t) + \frac{1}{2}\dot{x}^2(t)]dt$$

the boundary conditions are $x(1) = 4$, $x(t_f) = 4$, and $t_f > 1$ is free.
Does the extremal minimize or maximize the functional?

2. Find a curve that is an extremal for the functional

$$J(x) = \int_0^{t_f} \sqrt{1 + \dot{x}^2(t)} dt$$

t_f is free, $x(0) = 5$, and the end point must lie on the circle

$$x^2(t) + (t-5)^2 - 4 = 0$$

Verify your solution geometrically. Determine the final time.

3. Derive necessary conditions to find an admissible control $u^*(t)$ and an admissible trajectory $x^*(t)$ that make the performance measure extremum.

$$J(x) = h(\bar{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\bar{x}, \bar{u}, t) dt$$

$$\text{subject to } \dot{\bar{x}}(t) = \bar{f}(\bar{x}, \bar{u}, t)$$

$$\bar{x}(t_0) = \text{given},$$

$$t_f \text{ free}$$

$$\bar{x}(t_f) \text{ are on moving } \bar{\theta}(t) = 0$$

Explain how to solve the optimal problem.

4. Determine the $x(t)$ and the optimal control $u(t)$ to extremize the performance index

$$J = \frac{1}{2} \int_0^5 [x^2(t) + u^2(t)] dt$$

subject to $\dot{x}(t) = u(t)$, $x(0) = 2$, $x(5) = 0$.

<21년 10월 21일 12~14시 응시>

1번: 예제4.2.4

2번: 유제4.15

3번: 응용(노트62쪽의 general optimal control problem과 노트65쪽의 표를 같이 이용해 응용하기!)

4번: 노트58쪽의 5.1단원 예제